8.5: Regulation of enzyme activity helps control metabolism

Chemical chaos would result if all of a cell's metabolic pathways were operating simultaneously. Intrinsic to life's processes is a cell's ability to tightly regulate its metabolic pathways by controlling when and where its various enzymes are active. It does this either by switching on and off the genes that encode specific enzymes (which will be discussed in Concepts 18.1 and 18.2) or, as discussed here, by regulating the activity of enzymes once they are made.

Allosteric Regulation of Enzymes

In many cases, the molecules that naturally regulate enzyme activity in a cell behave something like reversible noncompetitive inhibitors (see Figure 8.18c): These regulatory molecules change an enzyme's shape and the functioning of its active site by binding to a site elsewhere on the molecule, via noncovalent interactions. **Allosteric regulation** is the term used to describe any case in which a protein's function at one site is affected by the binding of a regulatory molecule to a separate site. It may result in either inhibition or stimulation of an enzyme's activity.

Allosteric Activation and Inhibition

Most enzymes known to be allosterically regulated are constructed from two or more subunits, each composed of a polypeptide chain with its own active site. The entire complex oscillates between two different shapes, one catalytically active and the other inactive **(Figure 8.20a)**. In the simplest kind of allosteric regulation, an activating or inhibiting regulatory molecule binds to a regulatory site (sometimes called an allosteric site), often located where subunits join. The binding of an *activator* to a regulatory site stabilizes the shape that has functional active sites, whereas the binding of an *inhibitor* stabilizes the inactive form of the enzyme. The subunits of an allosteric enzyme fit together in such a way that a shape change in one subunit is transmitted to all others. Through this interaction of subunits, a single activator or inhibitor molecule that binds to one regulatory site will affect the active sites of all subunits.

Figure 8.20

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Allosteric regulation of enzyme activity.
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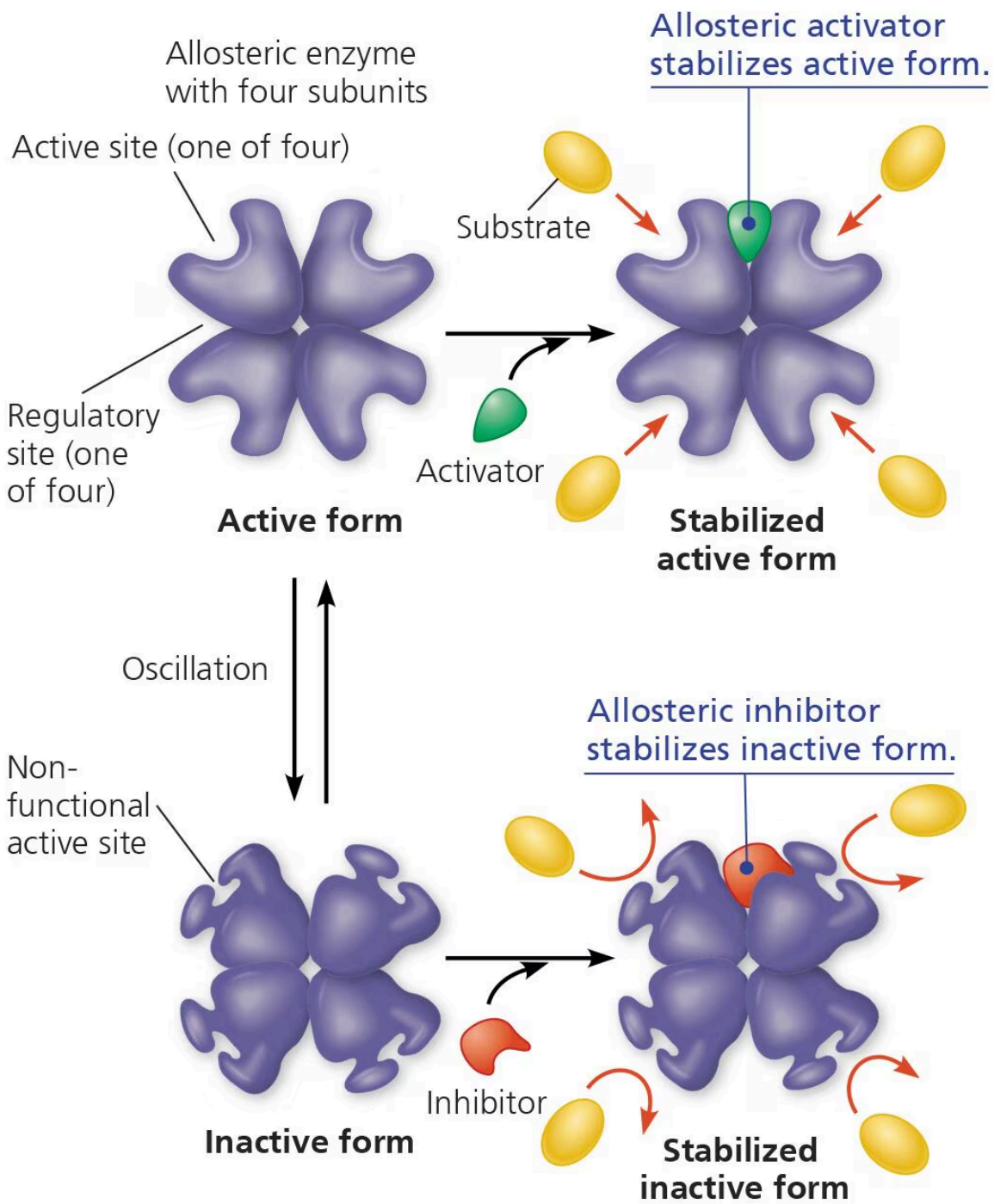
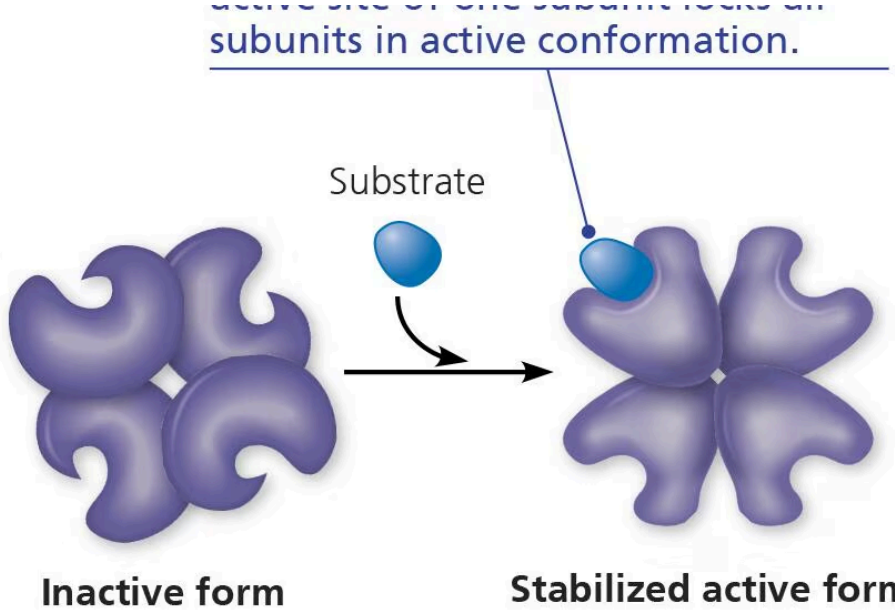
(a) Allosteric activators and inhibitors



At low concentrations, activators and inhibitors dissociate from the enzyme. The enzyme can then oscillate again.

(b) Cooperativity: another type of allosteric activation

Binding of one substrate molecule to active site of one subunit locks all



The inactive form shown on the left oscillates with the active form when the active form is not stabilized by substrate.

Part a. Allosteric activators and inhibitors: Active form of allosteric enzyme. Activator binds to regulatory site and stabilizes active form. In the inactive form, the inhibitor binds to regulatory site and stabilizes it. Active form of allosteric enzyme with four subunits, each with an active site and regulatory site between each subunit. The activator binds to the regulatory site and stabilizes the active form. The active form oscillates between the inactive form and the active sites become non-functional. The allosteric inhibitor binds to the regulatory site and stabilizes the inactive form. At low concentrations, activators and inhibitors dissociate from the enzyme. The enzyme can then oscillate again.

Part b. Cooperativity: Binding of substrate molecule to active site of subunit locks all subunits in active conformation and stabilizes active form. Inactive form oscillates with active form when active form is not stabilized.

Cooperativity: another type of allosteric activation. Binding of one substrate molecule to active site of one subunit locks all subunits in active conformation and stabilizes the active form. The inactive form oscillates with the active form when the active form is not stabilized by the substrate.

Fluctuating concentrations of regulators can cause a sophisticated pattern of response in the activity of cellular enzymes. The products of ATP hydrolysis (ADP and Ⓟi), for example, play a complex role in balancing the flow of traffic between anabolic and catabolic pathways by their effects on key enzymes. ATP binds to several catabolic enzymes allosterically, lowering their affinity for substrate and thus inhibiting their activity. ADP, however, functions as an activator of the same enzymes. This is logical because catabolism functions in regenerating ATP. If ATP production lags behind its use, ADP accumulates and activates the enzymes that speed up catabolism, producing more ATP. If the supply of ATP exceeds demand, then

catabolism slows down as ATP molecules accumulate and bind to the same enzymes, inhibiting them. (You'll see specific examples of this type of regulation when you learn about cellular respiration in the next chapter; see, for example, Figure 9.19.) ATP, ADP, and other related molecules also affect key enzymes in anabolic pathways. In this way, allosteric enzymes control the rates of important reactions in both sorts of metabolic pathways.

In another kind of allosteric activation, a *substrate* molecule binding to one active site in a multisubunit enzyme triggers a shape change in all the subunits, thereby increasing catalytic activity at the other active sites **(Figure 8.20b)**. Called **cooperativity**, this mechanism amplifies the response of enzymes to substrates: One substrate molecule primes an enzyme to act on additional substrate molecules more readily. Cooperativity is considered allosteric regulation because even though substrate is binding to an active site, its binding affects catalysis in another active site.

Although hemoglobin is not an enzyme (it carries O2 rather than catalyzing a reaction), classic studies of hemoglobin have elucidated the principle of cooperativity. Hemoglobin is made up of four subunits, each with an O2-binding site (see Figure 5.18). The binding of an O2 to one binding site increases the affinity for O2 of the remaining binding sites. Thus, where O2 is at a high level, such as in the lungs or gills, hemoglobin's affinity for O2 increases as more binding sites are filled. In O2-deprived tissues, however, the release of each O2 molecule decreases the O2 affinity of the other binding sites, resulting in the release of O2 where it is most needed. Cooperativity works similarly in multisubunit enzymes that have been studied.

Feedback Inhibition

Earlier, we looked at the allosteric inhibition of an enzyme in an ATP-generating pathway by ATP itself. This is a common mode of metabolic control, called **feedback inhibition**, in which a metabolic pathway is halted by the inhibitory binding of its end product to an enzyme that acts early in the pathway. **Figure 8.21** shows an example of feedback inhibition operating on an anabolic pathway. Some cells use this five-step pathway to synthesize the amino acid isoleucine from threonine, another amino acid. As isoleucine accumulates, it slows down its own synthesis by allosterically inhibiting the enzyme for the first step of the pathway. Feedback inhibition thereby prevents the cell from making more isoleucine than is necessary and thus wasting chemical resources.

Figure 8.21

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Feedback inhibition in isoleucine synthesis.
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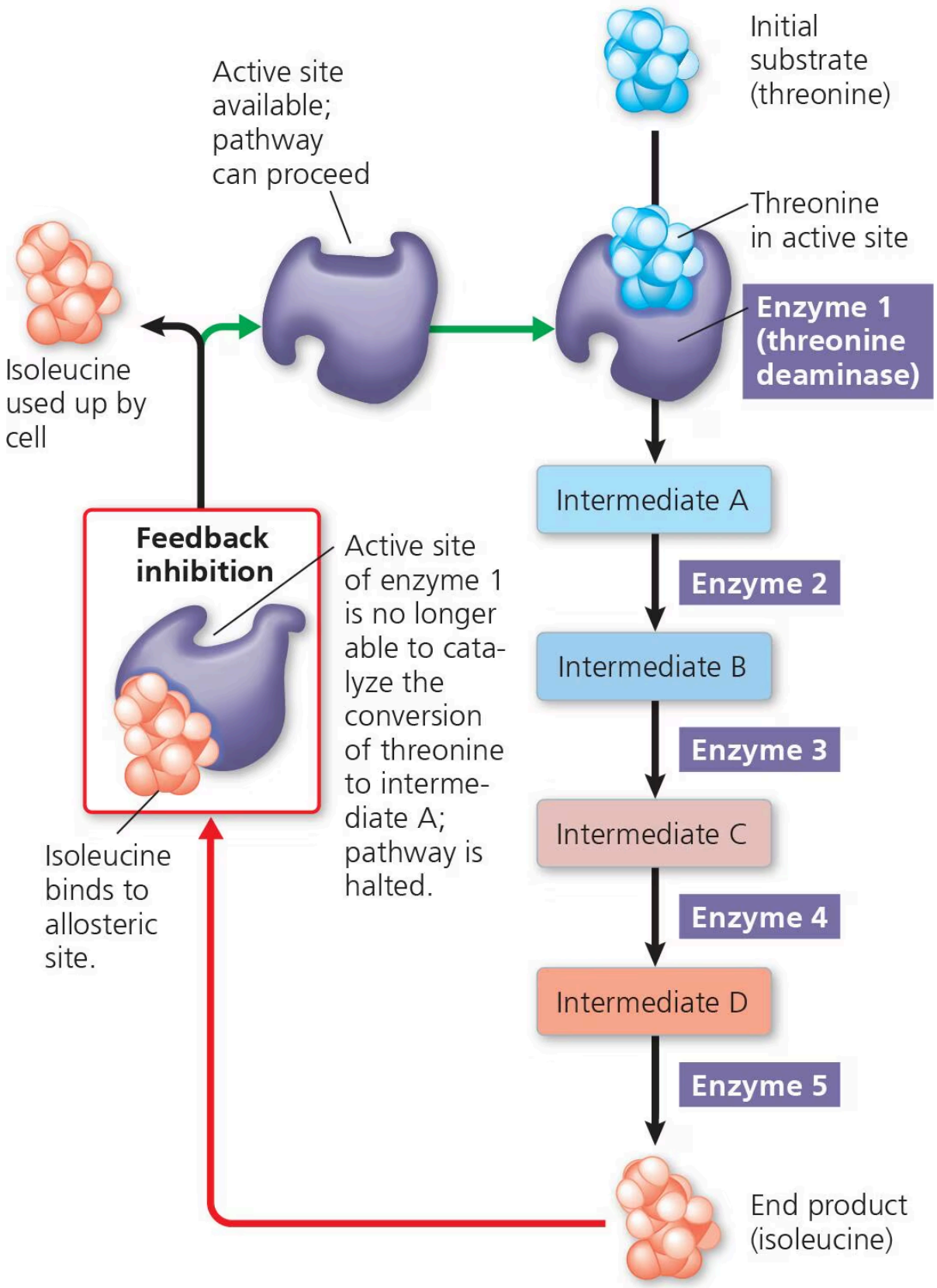
Initial substrate (threonine)
Active site available; pathway can proceed
Threonine in active site
Enzyme 1 (threonine deaminase)
Isoleucine used up by cell
Intermediate A
Feedback inhibition
Active site of enzyme 1 is no longer able to catalyze the conversion of threonine to intermediate A; pathway is halted.
Enzyme 2
Intermediate B
Enzyme 3
Intermediate C
Isoleucine binds to allosteric site.
Enzyme 4
Intermediate D
Enzyme 5
End product (isoleucine)

Threonine enters an anabolic pathway consisting of five catalyzed steps. The end product, isoleucine, allosterically inhibits the pathway in an example of metabolic control. The initial substrate threonine, an amino acid, binds to the active site of enzyme 1 (threonine deaminase). The product then enters a stepwise metabolic pathway where each intermediate product (A, B, C and D) are the results of reactions catalyzed by enzymes 2, 3 and 4. A final enzyme 5 catalyzes the reaction that converts intermediate D into the end product, isoleucine, also an amino acid. Isoleucine then binds to the allosteric site of enzyme 1, whose active site is no longer able to catalyze the conversion of threonine to intermediate A; the pathway is halted. As isoleucine is used up by the cell, the active site of enzyme 1 becomes available and the pathway can proceed.

Localization of Enzymes Within the Cell

The cell is not just a bag of chemicals with thousands of different kinds of enzymes and substrates in a random mix. The cell is compartmentalized, and cellular structures help bring order to metabolic pathways. In some cases, a team of enzymes for several steps of a metabolic pathway are assembled into a multienzyme complex. The arrangement facilitates the sequence of reactions, with the product from the first enzyme becoming the substrate for an adjacent enzyme in the complex, and so on, until the end product is released. Some enzymes and enzyme complexes have fixed locations within the cell and act as structural components of particular membranes. Others are in solution within particular membrane-enclosed eukaryotic organelles, each with its own internal chemical environment. For example, in eukaryotic cells, the enzymes for the second and third stages of cellular respiration reside in specific locations within mitochondria (**Figure 8.22**; see also Figure 6.32b).

Figure 8.22

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    Organelles and structural order in metabolism.
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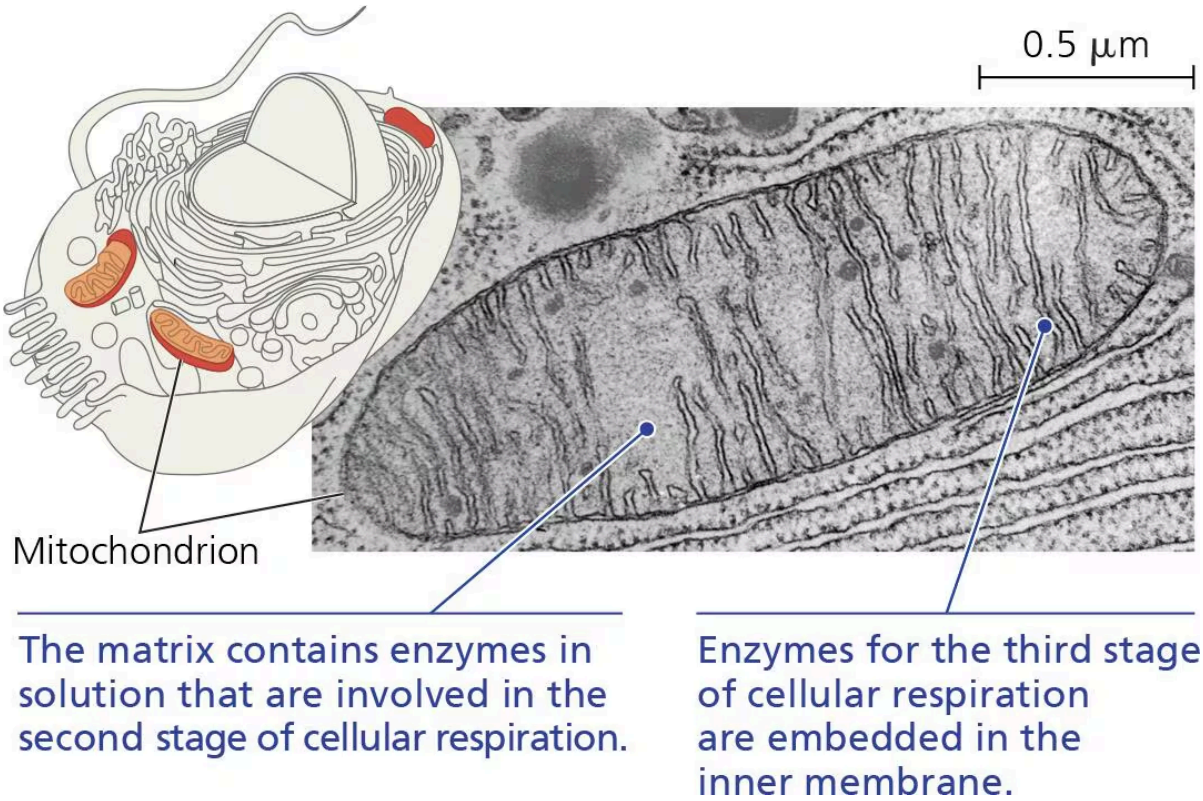

Mitochondrion shown on a scale of 1 micrometer. Mitochondria matrix contains enzymes in solution that are involved in one stage of cellular respiration. Enzymes for another stage of cellular respiration are embedded in the inner membrane.

Organelles such as the mitochondrion (TEM) contain enzymes that carry out specific functions, in this case the second and third stages of cellular respiration. (See also Figure 6.32b, lower left.)

In this chapter, you have learned about the laws of thermodynamics that govern metabolism, the intersecting set of chemical pathways characteristic of life. We have explored the bioenergetics of breaking down and building up biological molecules. To continue the theme of energy flow, and specifically of bioenergetics, we will next examine cellular respiration. This is the major catabolic pathway that breaks down organic molecules and releases energy that can be used for the crucial processes of life.